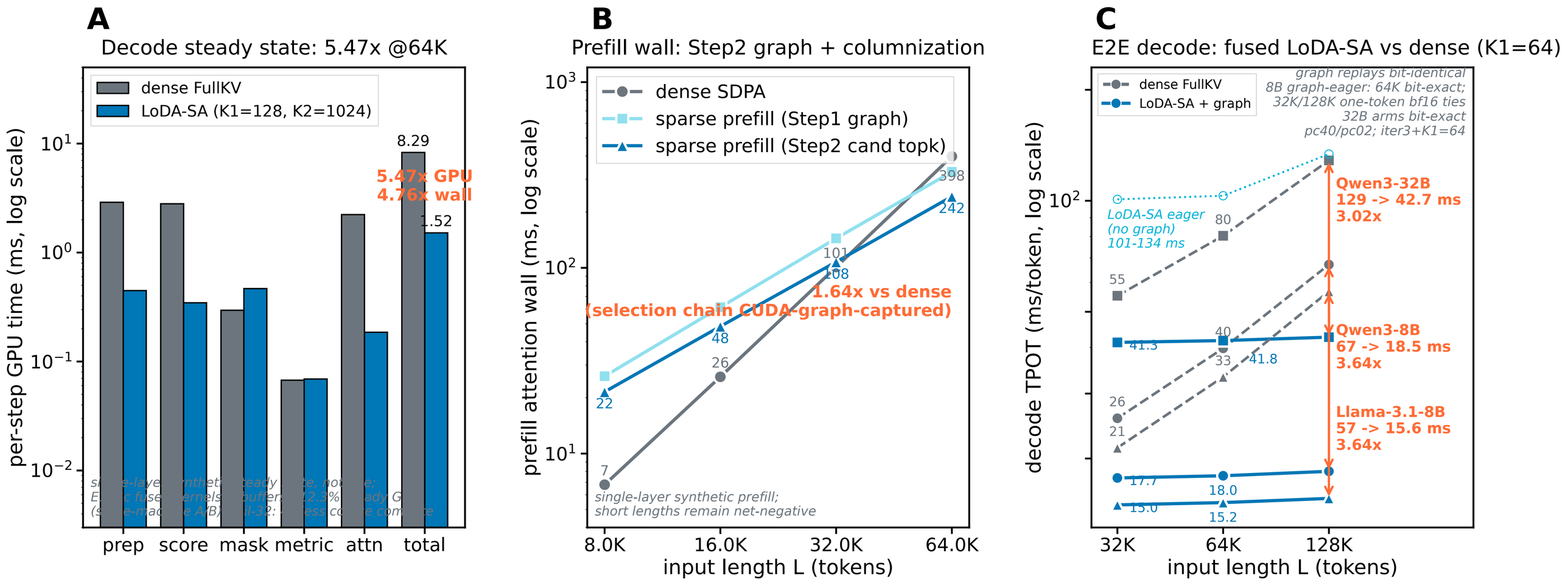
A
Decode steady state: 5.47x @64K
dense FullKV
LoDA-SA (K1=128, K2=1024)
per-step GPU time (ms, log scale)
8.29
1.52
5.47x GPU
4.76x wall
prep
score
mask
metric
attn
total
B
Prefill wall: Step2 graph + columnization
dense SDPA
sparse prefill (Step1 graph)
sparse prefill (Step2 cand topk)
prefill attention wall (ms, log scale)
398
242
101
108
48
22
26
7
1.64x vs dense
(selection chain CUDA-graph-captured)
single-layer synthetic prefill;
short lengths remain net-negative
8.0K
16.0K
32.0K
64.0K
input length L (tokens)
C
E2E decode: fused LoDA-SA vs dense (K1=64)
dense FullKV
LoDA-SA + graph
graph replays bit-identical
8B graph-eager: 64K bit-exact;
32K/128K one-token bf16 ties
32B arms bit-exact
pc40/pc02; iter3+K1=64
decode TPOT (ms/token, log scale)
LoDA-SA eager
(no graph)
101-134 ms
80
55
40
41.3
33
41.8
26
21
17.7
18.0
15.0
15.2
Qwen3-32B
129 -> 42.7 ms
3.02x
Qwen3-8B
67 -> 18.5 ms
3.64x
Llama-3.1-8B
57 -> 15.6 ms
3.64x
32K
64K
128K
input length L (tokens)